

age from malevolent network security attacks reached somewhere between \$157 billion and \$192 billion worldwide in 2004.

New threats – increase in operational complexity and manageability?

The current security defence paradigm is to deploy more and more of the existing security technologies throughout every segment of the network. This includes firewalls and ACLs to block access and perform application inspection, intrusion protection system (IPS) technology to provide very granular traffic inspection and identify known threats, encryption software to counter eavesdropping, anomaly detection to detect worms or DoS attacks, and antivirus software to battle viruses.

Many of today's security technologies were developed to perform their specific function with little context of the overall network threat environment. Operating alone, however, these technologies are less effective in stopping the newer attacks, as well as the changing ways in which user's access networks, because of the "security gaps" that exist between each technique's capabilities. With the increased complexity of threats, such as the blended threats that use a combination of techniques to disrupt networks, security technologies must operate in a coordinated fashion to stop attacks and better control network activity and applications.

Unfortunately, over the years, many companies have addressed nagging security concerns by constantly adding devices and software to address each particular problem. This has led to separate antivirus protection, firewalls, VPNs, and intrusion prevention. While this addresses the short-term needs, it creates an entirely new and bigger problem: managing multiple systems that operate independently of one another. As more advanced threats emerge, there is a need for network security to become more holistic; security technologies must act in coordination to detect and defend against more sophisticated threats. There is a growing need for devices that can assemble the puzzle pieces and lock down the gaps that exist in conventional network security systems.

Adaptive security for a changing world

Transforming chaos into a clear and manageable security policy is essential, which is why future network security systems need to focus on convergence and consolidation. For robust information security for an enterprise, a proactive architectural and system approach is critical. The idea is to accurately identify and stop attacks as early and as far from the destination host as possible, while simultaneously simplifying the security architectures required to do this.

Instead of a security product for each security need in isolation, an end-to-end security solution approach or system enables these combined functions to operate as a coordinated defence (instead of silos) that stops a broader range of attacks and greatly reduces the number of diverse multi-vendor devices that must be deployed, thereby simplifying security design and management.

Historically, firewalls have generally been considered fairly simple devices, but they are effective at what they do – either block a packet or let it through based on Layer 3 and Layer 4 information and session state. They can provide some level of application inspection but do not perform the detailed inspection of some other technologies. An IPS device can pick up where a traditional firewall leaves off by peering more deeply into a packet's contents to see whether the data within conforms to company policy. Host IPS, host firewall on servers and desktops/laptops, day zero protection and intelligent behavioral based protection from application vulnerability and related flaws (within or inserted by virus, worms or Trojans) provided great level of confidence on what is happening within an organisation on a normal day and when there is an attack situation, which segment and what has gone wrong and gives flexibility and control to stop such situations by having linkages of such devices with monitoring, log-analysis and event co-relation system.

Adaptive threat defence is the need now where an end-point security system can dynamically generate an attack signature and push it to the other end-points and to the perimeter IPS devices to stop it from propagating to rest of the infrastructure. Similarly security operations teams have challenges to fine-tune false positives, they struggle by logging into various devices to understand which logs are resulting into what logs; the need of the hour is that security operations team should be able to easily see the logs and link the same to configurations which are resulting into these logs.

This type of systems approach transforms security from operating as separate siloed technologies in a reactive mode-with limited and static detection methods-to functioning as a coordinated, proactive threat defense system that adapts to the threat environment.

These systems provide numerous benefits: improved detection, greater event classification accuracy, lower operating costs, streamlined administration, and services extensibility that integrate the most advanced security technologies as they are developed. These converged systems will not compromise the quality of security in any given category, but instead combine the strength of each in complementary ways to deliver a tighter, coordinated defence. ■

The author is Business Development Manager, Network Security, Cisco India & SAARC

Bomb alert!

A well-drafted model bomb threat plan, tested and evaluated with trained staff, will enable an organisation to promptly respond and help minimise loss

Raghu Raman

Warning: Terrorists will strike where it hurts the most

In his famous book *The Little Drummer Girl* John LeCarre wrote that the absolute terrorist is one who can blow up a room full of school children, without any remorse or hesitation. Abhorring as that might seem, the frightening recurrence with which such instances have occurred, necessitate corporate India to be prepared for the absolute terrorist.

Very recently, SIMI had threatened to blow up LIC buildings, a warning serious enough to mobilise the Mumbai police. In the past few months, many Indian companies in BFSI and IT / ITES sectors have been subject to hoax bomb threats. Though more often, the situation was brought under control – these incidents caused stress, confusion, disruption of work and loss of business and / or credibility. Inarguably, the problem merits attention. However, there is a need to remain calm and follow the basic rules of security. Employee awareness, training, practice and evacuation drills help to avert disasters and increase chances of correct response in times of crises.

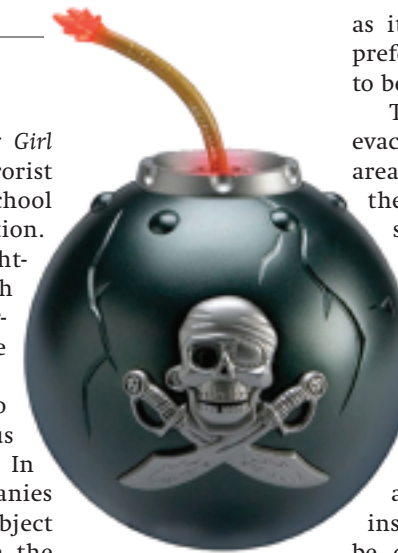
Dealing with bomb threats

A bomb threat may be received on telephone or through a written message. Experience shows that most of the threats received on telephones are hoaxes. However, till experts carry out a detailed search and come to that conclusion, the situation has to be taken seriously. Statistics suggest that one in a 1,000 calls may turn out to be genuine.

While administration and corporate security departments play a leading role in countering bomb threats, employees are often the first line of defence. A sound and practical bomb threat contingency plan ensures carrying out standardised drills and lays down specific responsibilities on all concerned.

Evacuation procedure

Once it is decided to evacuate, the security manager should give wide publicity by announcements through PA system, CCTV, megaphone, etc. Do not use the word "bomb"



as it may create panic and confusion. It is preferable to announce that "the area needs to be vacated – this is NOT a drill."

The announcement should specify the evacuation route, guide, location of assembly area and the need for staying there till further instructions. Route and assembly area should be searched before announcement.

Evacuation instructions (dos/don'ts):

Ideally the organisation should have practiced evacuation drills previously. If they have not—then these are some guidelines.

The announcer must remain calm, and advise the employees to follow the instructions. Similar announcements could be displayed on the monitors of CCTV, if

Security is a matter of trust.

- Software Protection *Sentry***
 - Anti Piracy
 - Web Authentication
 - Time Leasing
- Biometrics *BioSentry***
 - Time Attendance System
 - Physical Access Control
 - Logical Access Control (PC Logon)
 - Data Security
- Smart Card *SmartSentry***
 - Time Attendance System
 - Physical Access Control
- Software & Other Applications**
 - Time Attendance Software
 - Payroll Software
 - Canteen Management
 - Visitors Management
 - Library Management, etc

Wyse Biometrics Systems Pvt. Ltd.
 S. No. 82/1, Plot 20, Sahakar Nagar No. 1,
 Aranyashwar Road, Parvati, Pune.
 Tel.: 91-20 - 2422 4058, 24225288
 Email: sales@wyse.co.in
 Web: www.wyse.co.in

WYSE
 Since 1992

AN ELECTRONIC Identity Management ENTERPRISE

Software Protection
 Employee Training
 Fingerprint Biometrics
 Data Security

User Authentication
 Network Authentication
 Single Sign On
 Smart Cards

Physical Access Control
 Time Attendance
 Active Directory
 Access Control